



**School of Computer Science Engineering and Information
Systems
Fall Semester 2026-2027**

**Department of Computer Applications
PMCA698J – Dissertation -I
Review -1**

Date: 14.8.2026

**AI-Driven Blockchain Framework for Secure Medical Image Sharing
with Adaptive Chaos Encryption, Smart Access Control, and AI
Based Tamper Localization**

25MCA0261 - GOKUL S

Under the Guidance of

Ramesh P S

Assistant Professor Gr.2

SCORE

Guide Signature with date

**Internal Examiner -1
Signature**

**Internal Examiner-2
Signature**

ABSTRACT:

The proliferation of digital healthcare systems has created an urgent need for secure medical image exchange in hospitals, diagnostic centers, and medical professionals. Medical images like MRI, CT, X-ray, PET, and Ultrasound contain sensitive patient-specific data and are highly susceptible to unauthorized access, tampering, privacy breaches, and poor traceability in traditional centralized systems. This paper proposes a Blockchain Enabled Secure Medical Image Sharing Framework that leverages cryptography, blockchain technology, decentralized identifiers, smart access control, and explainable artificial intelligence for end-to-end security.

The proposed framework employs 4D Chen hyperchaotic permutation and diffusion alongside AES-256 encryption to secure uploaded medical images. Image fingerprints are generated using SHA-3 hashing, while encrypted images are stored in decentralized storage with their hash values stored in the blockchain for auditing and traceability. Permissions, access control lists, and smart contracts enforce role-based access while decentralized identifiers, verifiable credentials, and time-bound consents manage permissions. In case of tampering, a hybrid Swin-UNet architecture with SRM noise removal and attention mechanism localizes tampered regions while Grad-CAM highlights the image regions triggering model decisions. The paper proposes a secure medical image sharing framework that enhances confidentiality, integrity, authenticity, accountability, and traceability of medical images in a distributed blockchain environment for clinical applications.

Keywords: Blockchain, Medical Image Security, Hyperchaotic Encryption, AES-256, SHA-3, Smart Access Control, Decentralized Identity, Swin-UNet, Tamper Localization, Explainable AI.

[1.] INTRODUCTION:

The rapid growth of digital healthcare is revolutionizing the way medical information is created, stored, and shared across hospitals, diagnostic centers, clinicians, and other stakeholders. From MRI, CT, X-ray, PET, and ultrasounds scans, medical images are critical to analyzing a patient's condition and determining an appropriate treatment course. With the recent shift towards cloud- and network-based healthcare ecosystems, medical images are increasingly accessible and shareable across various nodes. At the same time, sensitive medical data becomes significantly exposed to the risks of exfiltration, unauthorized access,

manipulation, and forgery across distributed platforms.

This project addresses the problem of secure and private storage, sharing, and auditing of medical images in a distributed healthcare ecosystem. The medical imaging data is highly sensitive and should be protected from disclosure, unauthorized modification, and integrity violations. Typically, centralized storage and management of medical imaging data pose a challenge in terms of establishing trustworthy visibility into who accessed the data, when, and what changes were made. Conventional encryption methods can help secure medical images' confidentiality and integrity in digital storage, but they are usually insufficient in establishing cryptographically verified provenance of medical images. In addition, conventional access control methods usually offer limited flexibility in terms of fine-grained and decentralized permission management.

Leveraging the immutability of blockchain, this project proposes to store medical image fingerprints, permissions, and access control information as transactions in a blockchain to enable auditable provenance of medical images. By storing encrypted image fingerprints on the blockchain, medical image integrity can be routinely cryptographically verified. In addition, smart contracts can help manage permissions and regulate access to medical images in a decentralized manner. To this end, this project taps into Homomorphic encryption and decentralized storage to help secure sensitive medical image data in storage while being able to perform range queries on the encrypted data.

While the combination of encryption and blockchain can help secure the confidentiality and integrity of medical images, it is usually insufficient in determining the region of interest (RoI) where image forgery or tampering occurred. As such, this project also proposes leveraging artificial intelligence to determine RoI in medical images. Particularly, the project implements a hybrid architecture that combines a Swin-UNet and Vision Transformer with multi-scale residual modules for forgery detection and localization. Grad-CAM is also leveraged to help explain the model's decisions. The combination of conventional cryptography and machine learning can empower a distributed healthcare ecosystem to securely detect and localize tampering of medical images.

This project proposes an AI-Driven Blockchain Framework for Securing Medical Image Sharing with Adaptive Chaos Encryption, Smart Access Control, and AI-Based Tamper

Localization. It implements a hybrid medical image encryption framework that combines hyper chaos encryption and AES-256 to help ensure confidentiality and integrity of medical images in a distributed healthcare ecosystem. It also implements blockchain-based auditing and access control and uses AI-based tamper localization to help detect unauthorized modification of medical images. By combining confidentiality, integrity, access control, and auditability, the proposed solution seeks to facilitate secure and private medical image sharing in a distributed healthcare ecosystem.

[2.] PROBLEM STATEMENT:

Modern healthcare information management is undergoing a transition to the use of digital image storage and exchange infrastructure, which facilitates the sharing of MRI, CT, X-ray, PET, and ultrasound scans between hospitals and healthcare professionals. However, traditional medical image-sharing systems are challenged by the need for secure and private image storage and exchange. In particular, medical confidentiality may be compromised due to potential data breaches or unauthorized disclosure of personally identifiable health information, whereas image integrity can be impacted by the possibility of unauthorized image overwriting with no built-in means of detection and audit capabilities.

Confidentiality of medical images is typically addressed by conventional encryption algorithms and standardized access control mechanisms. At the same time, post-storage integrity protection remains a critical concern, as unauthorized alterations of medical images may occur at any point during their storage and exchange. Traditional blockchain technologies offer a potential solution to this problem, as they allow for the creation of an immutable log of image integrity checkpoints. Yet, in most cases, each hash value only indicates whether an image was altered in its entirety, whereas localized tampering typically requires additional integrity verification procedures. On the other hand, artificial intelligence-powered medical image tamper localization and detection solutions are typically hindered by the absence of an auditing infrastructure that would make tamper evidence tamper-proof. Furthermore, conventional access control mechanisms in medical image-sharing systems are often unable to accommodate decentralized user authentication, dynamically changing access control policies, auditable permission assignment, or fine-grained auditing of each image's sensitive information exchange.

Therefore, there is a need for a medical image-sharing framework that would address

confidentiality, integrity, access control, tamper localization and detection, and security auditing in a unified manner. The proposed project aims to design and implement a medical image-sharing framework that would be based on dynamic chaos encryption, AES-256, and SHA-3 standard encryption, blockchain-based integrity auditing, smart contract-based access control, decentralized identity management, and explainable AI-based medical image tamper detection to secure medical image exchange and ensure privacy-preserving auditing of medical image integrity and access authorization. The absence of confidentiality and integrity checkpoints in conventional medical image-sharing systems presents the primary challenge that the proposed project aims to address.

[3.] OBJECTIVES:

The general objective of this study is to design a secure and intelligent medical image sharing framework that leverages the potential of blockchain technology, cryptography, smart access control, and AI-assisted tamper localization.

1. Ensure the confidentiality of medical images using adaptive chaos-based encryption and AES-256 encryption algorithms to protect sensitive medical data from unauthorized access and disclosure
2. Guarantee the integrity of medical images using SHA-3 cryptographic hashing and blockchain-based digital watermarking for reliable verification of image authenticity and detection of any unauthorized modifications
3. Design confidential and decentralized medical image storage and sharing by adopting encrypted image storage with the assistance of content and transaction traceability provided by the blockchain
4. Enable image-level access control and auditing through the implementation of smart access control mechanisms
5. Localize image tampering by utilizing a Hybrid Swin-UNet model with SRM noise filtering and attention mechanisms for effective tamper detection and localization
6. Provide explainability for the tamper-localization algorithm using Grad-CAM to enable a visual interpretation of what the model is focusing on when detecting tampering
7. Maintain an immutable audit trail of image upload, access requests, permissions, integrity checks, and other security-related events using the capability of the blockchain
8. Combine cryptographic integrity verification with AI-assisted forensic analysis to allow the system to automatically perform tamper detection and localization when an integrity mismatch is detected.

9. Deliver an end-to-end secure medical image sharing framework that enhances the confidentiality, integrity, authenticity, accessibility, accountability, and traceability of medical images in a distributed medical image sharing platform.

[4.] SCOPE OF THE PROJECT:

The scope of this project is the implementation of a secure and intelligent framework for storage, verification, controlled access, and tamper analysis of medical images. The framework is intended to address the security requirements related to the sharing of medical images between hospitals, diagnostic centers, doctors, and other users.

1. Secure Medical Image Protection

The framework will be capable of storing and securing medical images of various types, including MRI, CT, X-ray, PET, and ultrasound scans. Medical images will be protected by chaos-based encryption/AES-256 to ensure confidentiality.

2. Blockchain-based Integrity Management

The blockchain will store cryptographic hashes and relevant transaction data to establish the verifiable integrity of medical images. This will help detect any unauthorized image alterations.

3. Decentralized and Secure Storage

The encrypted medical images can be stored on decentralized or IPFS-based storage, while the blockchain will store image integrity- and transaction-related information. This will help reduce the size of medical image data stored on the blockchain.

4. Smart Access Control

The framework will be equipped with smart access control mechanisms based on the role of the user, decentralized identities (DID), Verifiable Credentials, and time-bound permissions. This will help ensure that access to medical images is granted on a need-to-know basis.

5. AI-based Tamper Detection and Localization

The framework will utilize an AI-based tamper detection mechanism to localize regions of medical images that may have been altered. The mechanism will leverage the Hybrid Swin-UNet architecture with a SRM-based noise analysis and attention modules to detect image tampering.

6. Explainable Tamper Analysis

Explainable AI will be integrated to help blockchain auditors, medical professionals, and security analysts understand how the tamper detection model arrives at its conclusions. The

framework will leverage Grad-CAM to highlight image regions that were instrumental in the tamper detection model's final decision.

7. Immutable Security Auditing

The framework will allow for the auditing of image uploads, access, permission changes, integrity checks, and other sensitive activities. This will be achieved by leveraging the blockchain to store this information as transactions, thus making it difficult to alter or delete this data.

8. Integrated Security Workflow

The proposed project will deliver an integrated security framework for medical images that will combine cryptography, blockchain, access control, decentralized identifiers, and artificial intelligence to help secure medical images.

9. Future Healthcare Ecosystem Expansion

The framework will lay the foundation for the future expansion of a healthcare ecosystem that will involve hospitals, PACS, diagnostic centers, telemedicine, and other stakeholders.

Overall, the project scope will be focused on delivering a prototype of a secure medical image sharing framework that will address confidentiality, integrity, authenticity, access control, tamper localization, explainability, and auditing. The framework will be designed as a security and research prototype and not as a replacement for existing certified clinical diagnostic systems.

[4.] PROPOSED SYSTEM:

The proposed system refers to an AI Driven Blockchain Framework for Secure Medical Image Storage, Sharing, Access, and Tampering Localization with Chaos Based Encoding, Smart Access Control and Forensic Analysis. The system provides an integrated information security model based on cryptographic protections, blockchain, decentralized identifiers, smart access control, and AI forensic readiness to secure medical images.

The system is initiated by submitting the medical image for storage by the authorized healthcare user. The identity of the user (doctor, radiologist, administrator) is authenticated, whereas decentralized identifiers, verifiable credentials, and access control role-based policies manage permissions. Information related to patients and sensitive medical details is safeguarded by respective privacy and PHI protections.

A medical image is submitted to the system and subject to 4D Chen hyper chaotic encryption

algorithm including permutation and diffusion processes and AES 256 encryption to ensure confidentiality and protection of the image. In parallel, the SHA-3 cryptographic hash is generated from the image or its representation to capture its integrity description.

The medical image is securely stored (encrypted) in a storage layer, which can be IPFS-oriented. By contrast, the blockchain layer stores the cryptographic hash, timestamps, transactions, permission-related operations, and audit-related information about medical image storage and access. The absence of medical image files in blockchain layer is explained by the limitations in terms of space and capacity.

By accessing the medical image, users are subject to additional authentication and authorization measures, which are managed by smart contracts. Smart contracts also store and manage permissions for accessing the image, considering the role-based access control principles, verifiable credentials, and decentralized identifiers. In some cases, time-bound access permissions can be issued by the smart contracts to limit continuous access to the medical image. Activities conducted by users are recorded and stored to provide audit trails and support security assessments.

The process of verifying the integrity of a medical image involves recalculating its SHA-3 hash and comparing it with the hash stored in the blockchain, which represents the image's initial state. If hashes match, the image is considered intact; otherwise, the integrity check fails. In either case, the image is subject to additional forensic analysis. A hybrid Swin-UNET model with an SRM-based noise filter and attention mechanism is applied to identify image manipulations at multiple scales. The model provides a tamper localization map, indicating the regions in the image that were modified, whereas Grad-CAM explains the reasoning behind the results.

[5.] LITERATURE SURVEY:

S.NO	TITLE	MERITS	DEMERITS
1	Zero-Trust Medical Image Sharing: A Secure and Decentralized Approach Using Blockchain and the IPFS	This paper proposes a zero-trust architecture combining blockchain and IPFS for secure medical-image sharing. It improves decentralization, data integrity, and access security while reducing dependence on centralized storage.	The framework mainly concentrates on secure sharing and decentralized storage. It does not provide an advanced AI-based tamper localization mechanism to identify the exact manipulated region.
2	Decentralized Medical Image Sharing: A Blockchain Based Approach with Subject Sensitive Hashing for Enhanced Privacy and Integrity	The framework combines JPEG compression, feature extraction, encryption, IPFS, blockchain, and subject-sensitive hashing. Smart contracts are used to validate image-access requests, and the reported average correctness is 98%	The method focuses mainly on feature/hash-based integrity verification. It does not provide detailed AI forensic localization and explainability for manipulated regions.
3	Enhanced Security for Medical Images Using a New 5D Hyper Chaotic Map and Deep Learning Based Segmentation	A new 5D hyperchaotic map is combined with a customized U-Net for secure medical-image encryption. The method reports strong statistical-security results, including high entropy and low correlation	The approach mainly addresses image encryption. It does not integrate blockchain-based immutable auditing, decentralized identity, or smart access control into the complete sharing workflow.
4	An Encryption Algorithm for Multiple Medical Images Based on a Novel Chaotic System and an Odd-Even Separation Strategy	The paper proposes a novel chaotic system for encrypting multiple medical images and focuses on protecting sensitive medical information from unauthorized disclosure.	The work concentrates on cryptographic protection and does not provide an integrated blockchain framework or AI-based tamper localization.
5	Medical Image Encryption Based on Josephus Traversing and Hyperchaotic Lorenz System	The method combines Josephus scrambling, Arnold-map scrambling, hyperchaotic Lorenz sequences, and diffusion to protect medical images. It demonstrates resistance against common attacks and considers image damage during telemedicine.	The system focuses mainly on encryption and robustness. It does not provide blockchain-supported integrity records or intelligent localization of manipulated regions.

6	DICOM Compatible, 3D Multimodality Image Encryption Using Hyperchaotic Signal	The proposed method supports multiple medical-image modalities including MRI, CT, X-ray, and ultrasound. Hyperchaotic signals and multi-level diffusion are used to increase security, with an additional authentication mechanism.	The primary contribution is encryption and authentication. It does not combine encryption with blockchain auditability and AI-based forensic tamper localization.
7	A High-Security Chaotic Encryption Model for Biomedical Image Protection	The paper proposes a chaotic/hyperchaotic encryption approach based on an R3 model involving rotation, rescaling, and randomization. It aims to reduce pixel correlation and improve protection of biomedical images.	The method primarily focuses on confidentiality through encryption and does not address decentralized access control, blockchain verification, or AI-based tamper localization.
8	Novel Aspects of Secure Medical Image Encryption Using Block-Chain and Transformer-Based Deep Learning Model	This work combines blockchain technology with a Transformer-based deep-learning model to improve medical-image security, privacy, and integrity.	Although blockchain and deep learning are integrated, the approach does not fully address explainable tamper-region localization, decentralized identity, and time-limited access control as a unified framework.
9.	Building an IPFS and Blockchain-Based Decentralized Storage Model for Medical Imaging	The proposed framework uses IPFS and blockchain to reduce dependence on third-party centralized storage and provide a decentralized model for medical-image sharing.	The work primarily addresses decentralized storage and sharing. Advanced encryption, AI-based tamper detection, and explainable localization are not comprehensively integrated.
10	Blockchain-Enabled Security for Medical Image Transmission: Prescription Data Hiding and Multi-Secret Sharing-Based Encryption	This paper produces an automated system which predicts the heart disease accurately based on the symptoms according to gender/age and domain knowledge of experts in the field at the lowest cost.	Classification via clustering achieves poor compared to other two methods. Author intends to extend the work for predicting the intensity of the disease using fuzzy methods.
11	Swin-Unet: Unet-Like Pure Transformer for Medical Image Segmentation	Swin-UNet introduces a pure Transformer-based U-shaped architecture using shifted-window attention to capture both local and global features. It	The original model is designed for medical-image segmentation, not image-forensic tamper localization. It also does not address

		demonstrates strong performance in medical-image segmentation tasks.	blockchain, encryption, or secure image sharing.
12	TransForensics: Image Forgery Localization with Dense Self-Attention	TransForensics applies Transformer-based dense self-attention and correction modules for image-forgery localization. It can model global relationships and produce high-quality forgery masks	It is a general image-forensics approach rather than a complete medical-image security framework. It does not provide blockchain-based integrity evidence or secure medical-image access management.
13	TBFormer: Two-Branch Transformer for Image Forgery Localization	The patient and the doctor provided the project's input information. Heart illness is then analyzed using various AI algorithms using the doctor's inputs. The results are now compared to those of other models that have been used in the same area and are determined to be superior.	Accuracy cannot be guaranteed by a computerized system alone, and warehouse data is only as reliable as the data input that produce
14	Progressive Feedback-Enhanced Transformer for Image Forgery Localization	The ProFact model performs coarse-to-fine forgery localization by feeding localization information back into Transformer layers. It improves robustness and generalization across multiple forensic datasets.	The method is focused on general image-forgery localization. It does not connect the detected manipulation with blockchain-based integrity verification or medical-image access auditing.
15	Quantum-Inspired Post-Quantum Secure DCT Watermarking for Tamper Detection and Localization in Medical Images	This recent work proposes medical-image watermarking using SHA3-256-based keys, DCT embedding, tamper detection, and localization. It reports more than 80% tamper localization with high image quality and classification accuracy.	The approach is primarily watermark-based and does not provide a complete combination of blockchain-based sharing, smart access control, decentralized identity, hyperchaotic encryption, and explainable AI.

[6.] FINDINGS IN LITERATURE SURVEY:

The literature survey results demonstrate that the utilization of blockchain, cryptography, decentralized storage, access-control mechanisms, and artificial intelligence in medical image auditing and integrity checking have attracted researchers' attention. Blockchain-enabled approaches offer decentralization, integrity auditing transparency, and improved tamper

evidence, while IPFS and similar solutions tackle medical image storage challenges.

The literature survey also proves that encryption is pivotal to medical image security. However, conventional and advanced encryption algorithms, including chaotic and hyperchaotic ones, offer limited information about image tampering that may occur after image decryption, though they provide high confidentiality and resistance to unauthorized access. Additionally, it was found that blockchain-based medical image sharing systems enhance integrity, traceability, and access control to a greater extent compared to conventional approaches because of the ability to store hashes, permissions, medical image tags, and other verifiable attributes in an immutable manner.

While the reviewed research articles highlight the potential of utilizing IPFS decentralized storage in combination with blockchain to address the storage limitations of medical images, it was found that hashing functions only demonstrate that an image is altered when compared to the original one, without specifying the location of the alteration.

The literature survey also revealed that the implementation of access-control mechanisms, including smart contracts, attribute-based, and role-based access control, can help protect medical information and images from unauthorized access and ensure confidentiality and integrity. However, centralized access-control mechanisms are limited in terms of cross-institution auditing, traceability, and flexibility, and while decentralized solutions address some of these limitations, they introduce other challenges, including scalability.

The utilization of AI-based image-forensics approaches and models, including deep learning and transformers, to detect image tampering and localize image alterations is another research area that has garnered the attention of researchers. However, with few exceptions, these solutions are proposed as stand-alone image-forensics systems that are not designed to be combined with encryption, blockchain-based auditing, decentralized storage, identity verification, and access auditing, among others.

Therefore, the overall finding of the literature survey is that while the existing research articles address individual aspects of medical image security, there is a need to combine blockchain-based auditing, adaptive chaos-based encryption, AI-based localizing of image tampering, decentralized identity management, smart access control, and explainable AI to achieve a comprehensive, adaptive, and explainable medical image-sharing framework. In other words, the reviewed articles reveal a research gap that involves the design of an end-to-

end-to-end solution that can detect, localize, and provide forensic evidence of medical image tampering, among other benefits.

The proposed project would address this research gap by offering a security workflow in which blockchain-enabled hash verification would be utilized to detect a tampering attempt, after which an AI-based forensic module would be utilized to localize the suspicious region and explain its presence to a user.

Key Findings

1. Blockchain enhances integrity auditing and traceability in medical image sharing but cannot localize image tampering.
2. Encryption mechanisms offer confidentiality but do not provide post-storage tamper evidence.
3. IPFS decentralized storage addresses the problem of limited storage space but requires additional measures to enhance security and integrity.
4. Smart access control mechanisms do not scale well and offer limited flexibility in managing authorization across institutions.
5. AI-based image-forensics models are proposed as stand-alone systems and are limited in terms of combining them with blockchain-based auditing, decentralized storage, and encryption mechanisms.
6. When designing AI-based image tampering localization systems, it is crucial to consider the explainability of the results because it assists users in identifying the image regions that influenced the outcome of the tampering detection process.
7. Combining blockchain-based integrity verification tools and AI-based localization mechanisms can address more aspects of medical image sharing security, including ensuring tamper evidence and providing localization of the suspicious regions.
8. Overall, the literature survey results demonstrate that medical image-sharing platforms need an end-to-end security solution that offers forensic evidence, auditability, integrity, confidentiality, and flexibility, which are not addressed by the utilization of one security mechanism.

[8.] METHEDOLOGY:

The proposed system follows a multi-layer methodology that integrates medical image encryption, blockchain-based integrity verification, decentralized storage, smart access control, and AI-based tamper localization. The methodology is designed to protect medical images throughout their lifecycle, from upload and encryption to authorized access, integrity verification, tamper analysis, and auditing.

8.1 Medical Image Acquisition

Medical images such as MRI, CT, X-ray, PET, and ultrasound images are collected from suitable medical-image datasets or authorized healthcare sources. The images are preprocessed into a standardized format and resolution where required. Patient-related information and sensitive metadata are handled separately to minimize unnecessary exposure of Protected Health Information (PHI).

8.2 User Authentication and Identity Management

Before uploading or accessing an image, the user is authenticated through secure identity-management mechanisms. Decentralized Identifiers (DID) and Verifiable Credentials (VC) are used to represent and validate user identity. Multi-factor authentication can additionally be applied to strengthen account security.

The system assigns users appropriate roles, such as:

- Doctor
- Radiologist
- Hospital Administrator
- Researcher
- Patient

These roles are subsequently used by the access-control layer.

8.3 Smart Access Control

A Role-Based Access Control (RBAC) mechanism is implemented to determine whether a user is authorized to access a particular medical image. The framework can additionally support time-limited permissions, allowing access to be granted only for a specified period.

The access-control process evaluates:

User Identity → Credentials → User Role → Permission → Validity Period → Access Decision

8.4 Medical Image Encryption

After successful authentication, the medical image is protected using a multi-layer encryption process. The proposed framework employs a 4D Chen hyperchaotic encryption mechanism involving permutation and diffusion operations.

The general process is:

Original Image → Hyperchaotic Key Generation → Pixel Permutation → Pixel Diffusion → AES-256 Encryption → Encrypted Image

The hyperchaotic process increases the randomness and reduces the statistical relationship between the original and encrypted images. AES-256 provides an additional standardized cryptographic protection layer.

8.5 SHA-3 Integrity Generation

After encryption, a SHA-3 cryptographic hash is generated for the protected image or the selected integrity representation.

The generated hash acts as a unique fingerprint:

Encrypted Image → SHA-3 → Integrity Hash

Even a small modification to the protected image should result in a different hash value. This hash is used later for integrity verification.

8.6 Decentralized Storage

The encrypted medical image is stored in a secure storage layer, such as an IPFS-oriented decentralized storage system. Instead of storing the complete medical image directly on the blockchain, the blockchain maintains the associated hash, content identifier, timestamp, and transaction information.

This approach reduces blockchain storage requirements while maintaining verifiable evidence of the stored image.

8.7 Blockchain Registration

A blockchain transaction is generated when a medical image is registered. The transaction can contain:

- Image hash
- Content identifier
- Upload timestamp
- User/owner identifier
- Permission information
- Transaction identifier
- Integrity status

Because blockchain records are tamper-evident and distributed, they provide a reliable reference for subsequent verification and auditing.

8.8 Secure Medical Image Sharing

When an authorized healthcare professional requests an image, the system verifies the user's identity and access permissions before allowing retrieval.

The process is:

Access Request → Identity Verification → RBAC Verification → Permission/Time Validation → Blockchain Verification → Image Retrieval

Unauthorized users are denied access, while authorized operations are recorded in the audit trail.

8.9 Integrity Verification

When the encrypted image is retrieved, its current integrity hash is calculated again using SHA-3.

The system compares:

Current Hash $\neq$ Blockchain Hash $\rightarrow$ Possible Modification

Current Hash = Blockchain Hash $\rightarrow$ Integrity Verified

This mechanism provides the first level of tamper detection.

8.10 AI-Based Tamper Localization

A hash mismatch indicates that the image has changed, but it does not identify **where** the modification occurred. Therefore, the proposed framework introduces an AI-based forensic layer.

A **Hybrid Swin-UNet model** is used for pixel-level tamper localization. The architecture combines:

- **Swin Transformer** for global and local feature extraction
- **UNet-style encoder-decoder architecture** for precise localization
- **SRM filtering** for extracting manipulation-related noise features
- **Attention mechanisms** for emphasizing suspicious regions

The model produces a **tamper localization mask** identifying potentially manipulated areas.

8.11 Explainable AI Using Grad-CAM

To improve interpretability, **Grad-CAM** is used to visualize the image regions that contributed significantly to the model's prediction.

The output provides:

Medical Image → AI Prediction → Grad-CAM Heatmap → Suspicious Region Visualization

This helps users understand the AI decision instead of relying only on a binary tampered/non-tampered result.

8.12 Blockchain-Based Audit Trail

Security-related activities are recorded to provide traceability. These activities may include:

- Image upload
- Image encryption
- Image registration
- Access request
- Access approval/rejection
- Permission modification
- Integrity verification
- Tamper detection
- Image sharing

The audit records provide evidence of **who performed an operation, what operation was performed, and when it occurred.**

8.13 Performance and Security Evaluation

The proposed framework is evaluated using both security and AI-performance metrics.

Encryption evaluation:

- Entropy
- Correlation coefficient
- NPCR
- UACI
- Key sensitivity
- Encryption/decryption time

AI evaluation:

- Accuracy
- Precision
- Recall
- F1-score
- IoU
- Dice coefficient

System evaluation:

- Blockchain transaction latency
- Image upload/retrieval time
- Access-control response time
- Storage overhead
- Hash-verification time

The results are compared with suitable baseline methods to determine whether the integrated framework provides improvements in security, tamper localization, explainability, and operational efficiency.

[9.] SOFTWARE REQUIREMENTS:

[9.1] Functional Requirements

The functional requirements specify the processes and services that the proposed AI-Driven Blockchain Framework for Secure Medical Image Sharing must deliver to the users and/or administrators.

9.1.1 User Registration and Authentication

- The proposed system shall support user registration and authentication.
- The system shall perform user authentication prior to allowing users access to the medical-image services.
- The system shall provide role-based user identification.
- The system shall offer multifactor authentication (if needed).

- The system shall support decentralized identifiers (DID)/verifiable credentials (VC).

9.1.2 Medical Image Upload

- The proposed system shall allow users to upload medical images.
- The system shall support the upload of various medical image formats including MRI, CT, X-ray, PET, and ultra-sonography.
- Prior to processing, the system shall verify the uploaded file.
- The system shall ensure that the uploaded image is associated with the pertinent patient and metadata.
- The system shall disallow unauthorized users from uploading or registering medical images.

9.1.3 Medical Image Encryption

- The proposed system shall encrypt medical images.
- To protect the medical images, the system shall adopt 4D chen hyperchaotic permutation and diffusion.
- The system shall use AES-256 encryption to encrypt the data.
- The system shall protect the encryption parameters/keys.
- The system shall restrict decryption privileges to authorized users.

9.1.4 Integrity Hash Generation

- The proposed system shall generate a SHA-3 cryptographic hash of the protected image.
- The hash value shall be linked to the protected image.
- The system shall use the hash value to establish integrity of the retrieved image.
- The system shall use the hash to compare against the blockchain-recorded hash.

9.1.5 Blockchain Registration

- The proposed system shall register pertinent image information in the blockchain.
- The information stored in the blockchain shall include the image hash and other transaction information.
- The system shall store relevant timestamps in the blockchain.
- The system shall protect the integrity of the stored records in the blockchain.
- The system shall prevent unauthorized modification of the stored records in the blockchain.

9.1.6 Secure/Decentralized Image Storage

- The proposed system shall securely store encrypted medical images in a storage layer.
- The system shall support decentralized or IPFS-based storage.
- The system shall ensure the association of a stored image with its blockchain record.
- The system shall restrict access to stored images to authorized users.

9.1.7 Smart Access Control

- The proposed system shall enforce role-based access control (RBAC).
- The system shall enforce access control policies based on the user's roles.
- The system shall allow access to resources only after successful user authentication.
- The system shall support timed permissions.
- The system shall allow the administrator or an authorized user to revoke permissions.
- The system shall deny unauthorized access requests.

9.1.8 Medical Image Sharing

- The proposed system shall allow authorized healthcare users to request medical images.

- Before granting access to medical images, the system shall authenticate the requesting user.
- The system shall share the medical images only with the requesting users who satisfy the system's security policy.
- The system shall record image-sharing events for auditing purposes.
- The system shall prevent unauthorized users from downloading or viewing medical images.

9.1.9 Integrity Verification

- The proposed system shall verify the integrity of medical images.
- When retrieving medical images or conducting integrity checks, the system shall re-compute the image hash.
- The system shall compare the newly generated hash with the stored hash.
- The system shall report the verification result.
- When the two hashes do not match, the system shall flag the image for further investigation.
- If the system detects possible image tampering, it shall conduct a forensic analysis.

9.1.10 AI-Based Tamper Detection and Localization

- The proposed system shall allow AI-based medical-image tamper analysis.
- The system shall use the Hybrid Swin-UNet architecture to localize image tampering.
- The system shall use SRM-based noise analysis to extract image manipulation features.
- The system shall use attention mechanism to enhance tamper localization.
- The system shall provide a mask of the suspicious image region.
- The system shall differentiate between verified and tampered medical images.

[9.2] Non Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics that the proposed AI-Driven Blockchain Framework for Secure Medical Image Sharing must satisfy. These requirements ensure that the system is secure, reliable, efficient, scalable, and easy to use.

9.2.1 Security

- The system shall protect medical images against unauthorized access, modification, and disclosure.
- Sensitive medical images shall be encrypted using strong cryptographic mechanisms.
- Blockchain records shall provide tamper-resistant integrity and audit information.
- Authentication and access-control mechanisms shall prevent unauthorized users from accessing protected resources.
- Encryption keys and authentication credentials shall be securely managed.

9.2.2 Confidentiality

- Patient information and medical images shall remain confidential during storage and transmission.
- Only authenticated and authorized users shall be able to access protected medical images.
- Sensitive metadata and PHI shall not be unnecessarily exposed on the blockchain.
- The system shall avoid storing complete medical images directly on the blockchain.

9.2.3 Integrity

- The system shall maintain reliable integrity verification for stored and shared medical images.
- SHA-3 hashes shall provide cryptographic references for detecting unauthorized modifications.

- Blockchain records shall provide an immutable reference for registered image information.
- Any detected inconsistency shall be reported to the appropriate user or administrator.

9.2.4 Performance

- Image encryption and decryption should be completed within an acceptable processing time.
- Hash generation and integrity verification should introduce minimal delay.
- Blockchain transactions should be processed within an acceptable response time.
- AI-based tamper localization should provide results without excessive computational delay.
- The system should efficiently handle different medical-image sizes and formats.

9.2.5 Reliability

- The system shall operate consistently during image upload, encryption, storage, retrieval, and verification.
- Medical images shall not be corrupted during encryption, transmission, or decryption.
- Failed transactions or interrupted operations shall be handled safely.
- The system should maintain consistent records between the storage layer and blockchain.

9.2.6 Availability

- Authorized users should be able to access the system whenever the required services are operational.
- The storage layer should provide reliable availability of encrypted medical images.
- Blockchain records should remain accessible for integrity verification and auditing.
- System failures should not result in unauthorized exposure of medical information.

[10.] SYSTEM ARCHITECTURE:

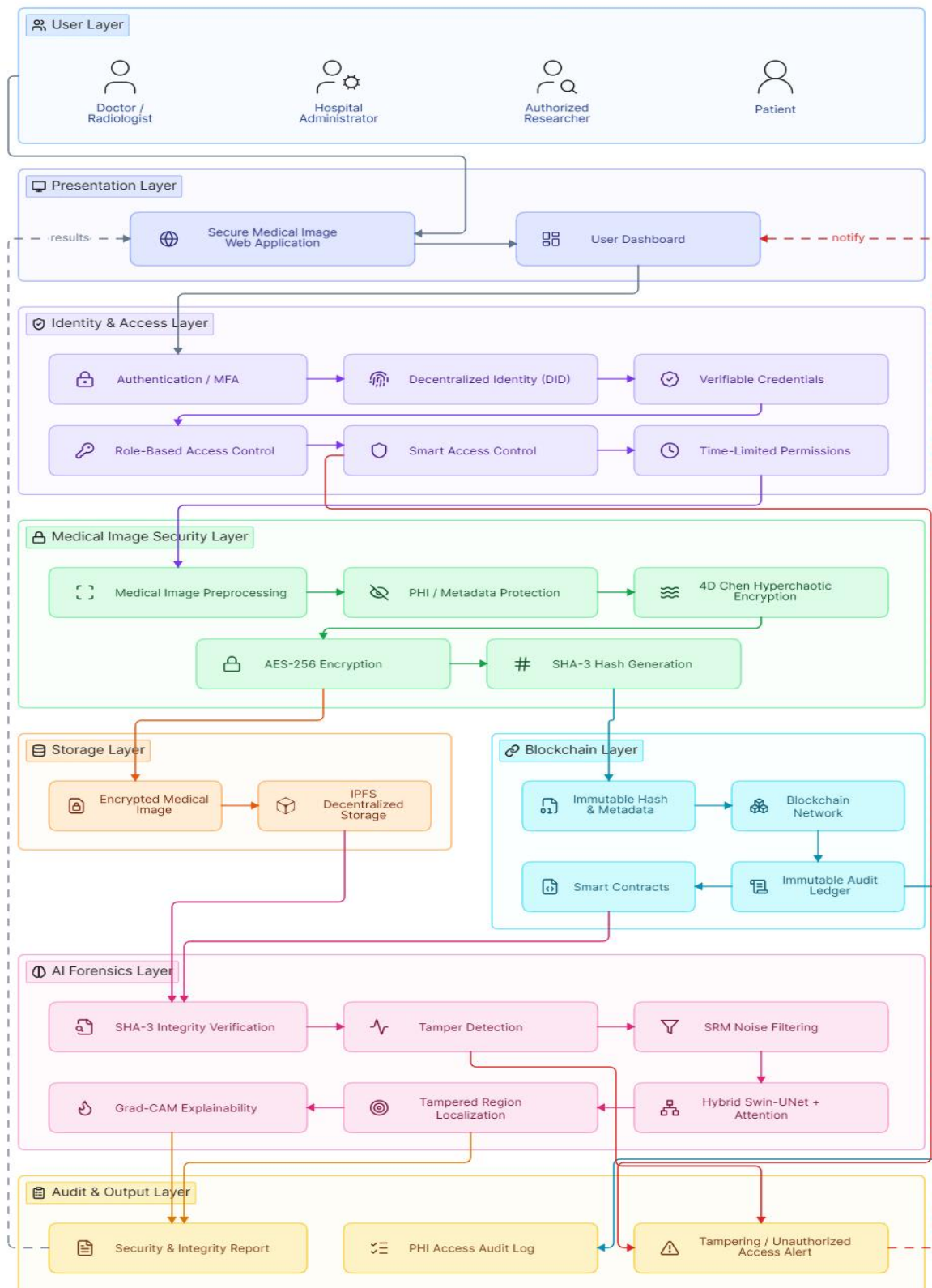
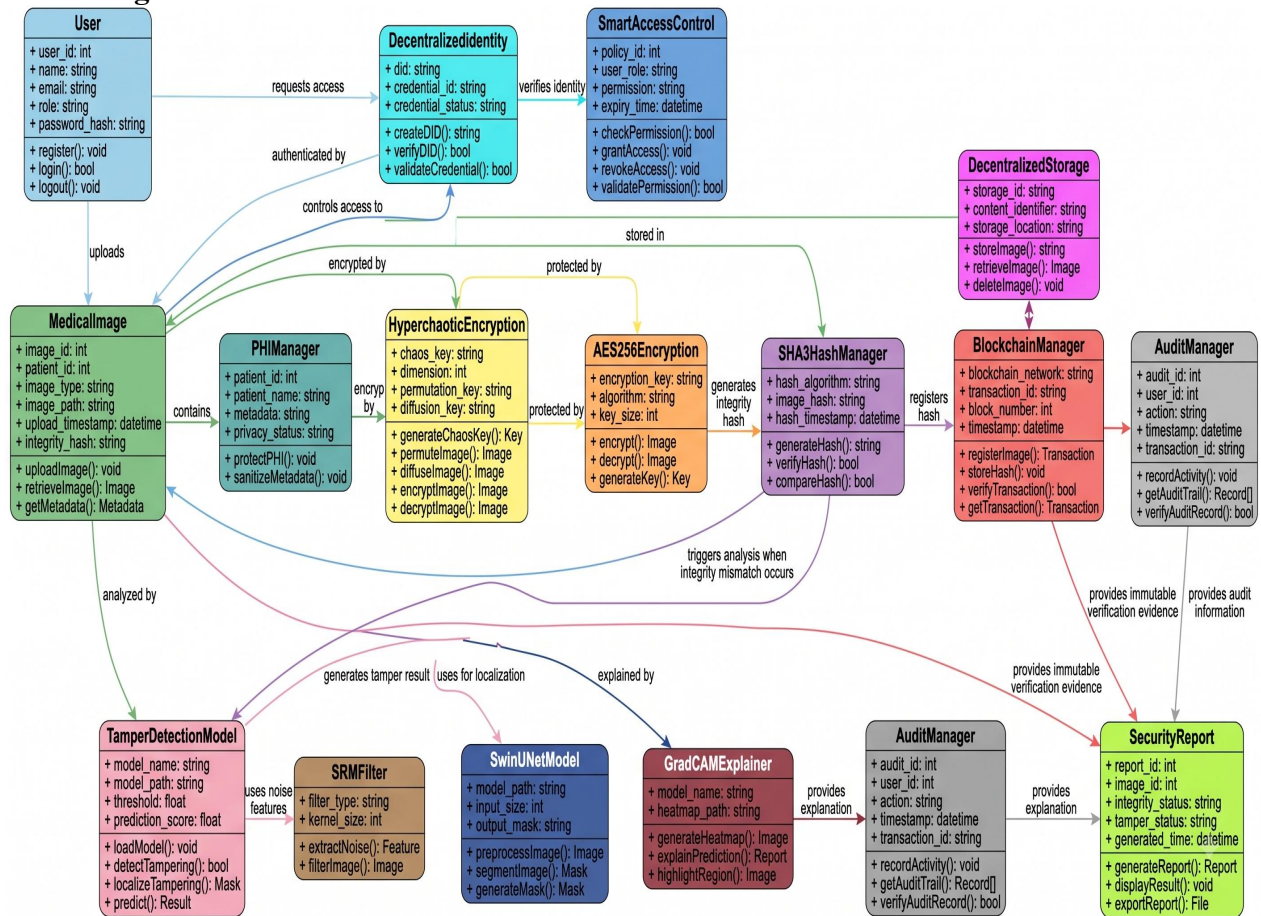


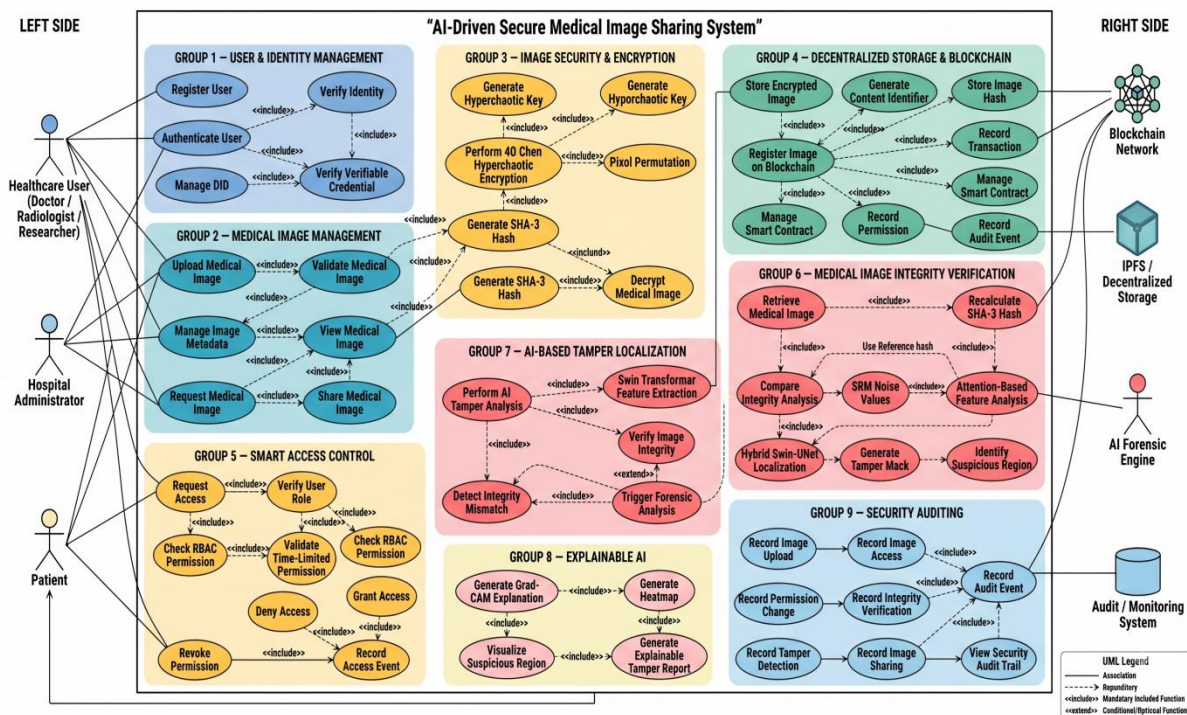
Fig. 10.1 System Architecture

[10]. UML DIAGRAMS:

Class Diagram :



Use Case



[11]SUMMARY:

The proposed project, “AI-Driven Blockchain Framework for Secure Medical Image Sharing with Adaptive Chaos Encryption, Smart Access Control, and AI-Based Tamper Localization,” aims to develop a secure and intelligent framework for protecting medical images such as MRI, CT, X-ray, PET, and ultrasound scans during storage and sharing. The framework combines 4D Chen hyperchaotic encryption, AES-256 encryption, SHA-3 hashing, blockchain technology, decentralized identity, smart access control, and explainable artificial intelligence to address the major security challenges in digital healthcare.

The proposed system protects medical images from unauthorized access and disclosure through hybrid encryption while blockchain provides immutable records of image hashes, permissions, transactions, and access activities. Role-Based Access Control (RBAC), Decentralized Identifiers (DIDs), Verifiable Credentials, and time-limited permissions provide secure and controlled image access. In addition, a Hybrid Swin-UNet model with SRM noise filtering and attention mechanisms is used to detect and localize suspicious modifications in medical images. Grad-CAM provides visual explanations of the regions identified as potentially tampered.

Overall, the proposed framework integrates confidentiality, integrity, authentication, access control, tamper detection, explainability, auditing, and traceability into a unified medical-image sharing system. By combining blockchain, advanced cryptography, decentralized access management, and AI-based forensic analysis, the system is intended to provide a more secure, transparent, and trustworthy solution for medical image exchange in distributed healthcare environments.

REFERENCES:

- [1] Arafat, Y., Siam, A. S. M., Chowdhury, M. M., Hasan, M. M., Jobayer, S. H., Shatabda, S., Islam, S., & Mukta, S. (2025). Decentralized medical image sharing: A blockchain-based approach with subject sensitive hashing for enhanced privacy and integrity. *IET Blockchain*, 5(1), e70009.
- [2] Archana, G., Goyal, R., & Kumar, K. M. V. M. (2025). Blockchain-driven optimized chaotic encryption scheme for medical image transmission in IoT-edge environment. *International Journal of Computational Intelligence Systems*, 18, 11.

- [3] Amdouni, R., Madani, M., Hajjaji, M. A., Bourennane, E. B., & Atri, M. (2025). Secure medical image transmission using chaotic encryption and blockchain-based integrity verification. *Computers, Materials & Continua*, 84(3), 5527–5553.
- [4] Shahid, U., Kanwal, S., Bano, M., Inam, S., Abdalla, M. E. M., & Shaikh, Z. A. (2025). Blockchain driven medical image encryption employing chaotic tent map in cloud computing. *Scientific Reports*, 15, 6236.
- [5] Afzal, S., Bokhari, M. U., Khan, I., Siddiqui, M. A., Ahmad, A., & Aijaz, M. (2025). A hybrid encryption model for medical image security in IoT healthcare. *Discover Computing*, 28, 321.
- [6] Mallick, S. R., Sobhanayak, S., & Lenka, R. K. (2025). Secure and trusted data sharing in smart healthcare using blockchain and IoT integration. *Discover Internet of Things*, 5, 90.
- [7] Zhao, A., & Tian, H. (2025). Secure sharing of electronic medical records based on blockchain and searchable encryption. *Electronics*, 14(13), 2679.
- [8] Mahmood, S. D., Drira, F., Mahdi, H. F., & Alimi, A. M. (2025). Secure medical image sharing: Technologies, watermarking insights, and open issues. *IEEE Access*.
- [9] Li, M., Rachana, H. A., Yu, C., Yu, Y., Zhong, R. R., & Huang, G. Q. (2025). Blockchain-based medical data asset sharing framework for Healthcare 4.0. *IEEE Transactions on Industrial Informatics*, 21(4), 2779–2788.
- [10] Man, Z., & Li, S. (2026). Data protection and efficient transmission for smart healthcare: An RDHEI approach based on LZW compression and composite key mechanism. *Expert Systems with Applications*, 296, 129040.